Predicting Familial Hypercholesterolemia

Using Genomic Variant Expression Data

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Agenda

1. Who gets Familial Hypercholesterolemia (FH)?
2. Which Gene Variants Drive FH?
3. Can we predict FH?
4. Limitations

Who gets FH?

2,761

Total individuals

in the dataset

2,213

FH cases

80% of dataset

265

Genetic variant features

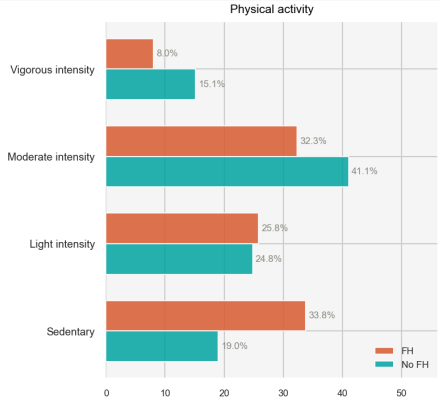
Var_X100000–Var_X100264

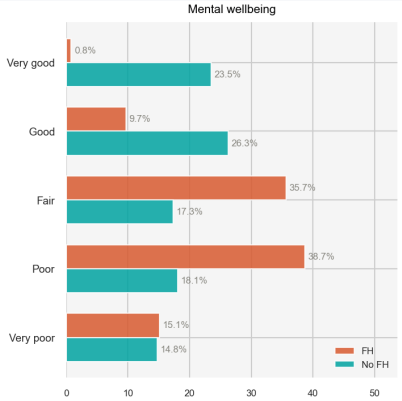
DATASET VS UK POPULATION

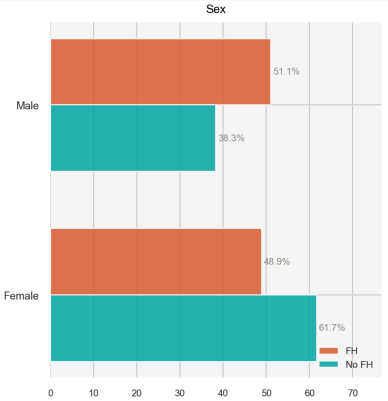
1 in 250

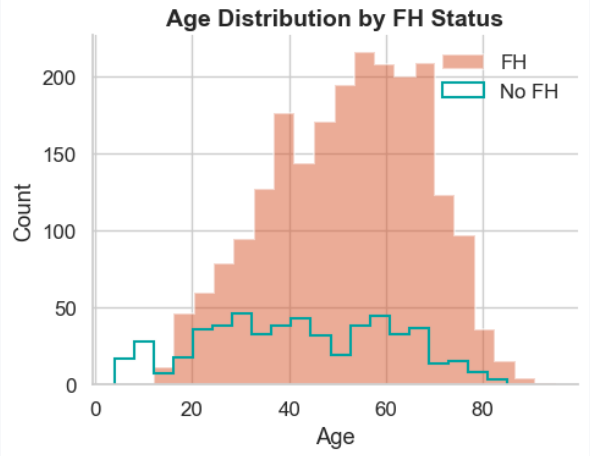
UK population prevalence (0.4%)

Currently identified < 8%







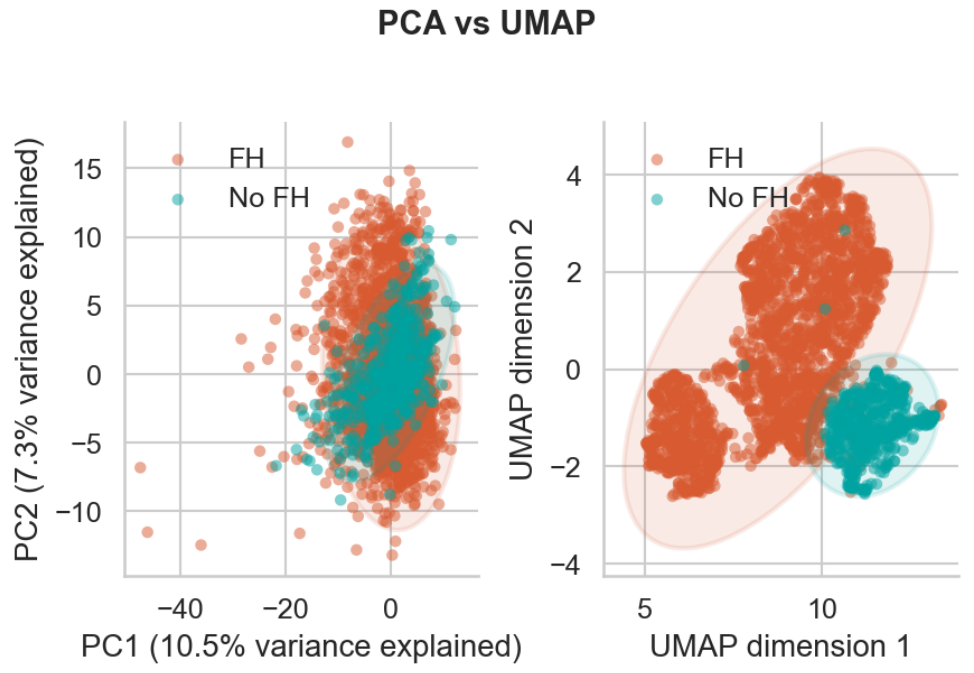


Who gets FH?

How demographic and self-reported health measures correlate with FH status

Logistic Regression Variable	Odds Ratio	95% CI	p-value
Age	1.04	(1.03, 1.05)	< 0.001
Sex: Female (ref)	1.00	—	—
Sex: Male	1.86	(1.48, 2.33)	< 0.001
Physical Activity: Sedentary (ref)	1.00	—	—
Physical Activity: Light	0.53	(0.38, 0.74)	< 0.001
Physical Activity: Moderate	0.34	(0.25, 0.47)	< 0.001
Physical Activity: Vigorous	0.42	(0.28, 0.65)	< 0.001
Mental Wellbeing: Very Poor (ref)	1.00	—	—
Mental Wellbeing: Poor	2.29	(1.63, 3.22)	< 0.001
Mental Wellbeing: Fair	2.03	(1.43, 2.88)	< 0.001
Mental Wellbeing: Good	0.35	(0.24, 0.49)	< 0.001
Mental Wellbeing: Very Good	0.03	(0.02, 0.05)	< 0.001

Which Gene Variants Drive FH?



Dimensionality Reduction

Key findings

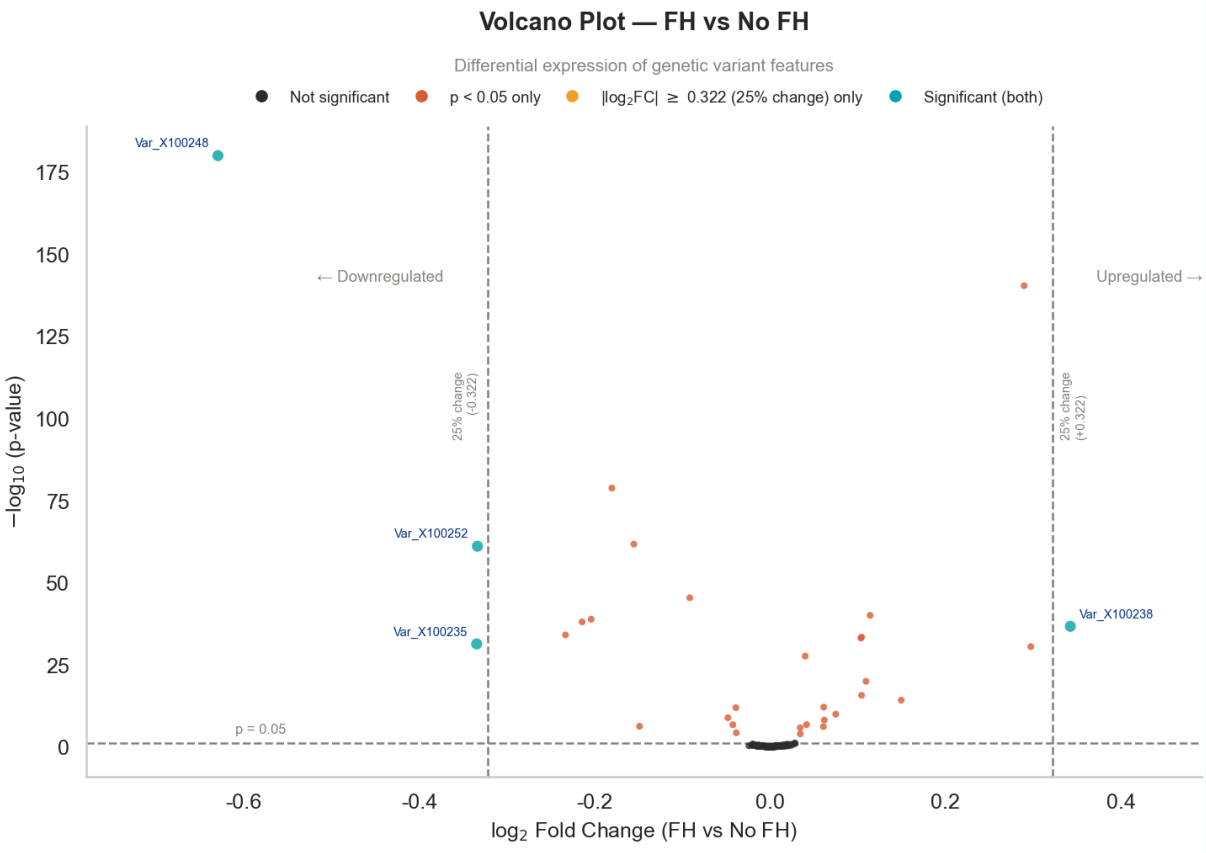
Signal is spread across many components — no single dominant gene driving FH

No clear cluster separation in PCA under linear projection

UMAP captures non-linear structure beyond PCA's reach

Clearer FH vs No FH cluster separation visible in UMAP — consistent with polygenic architecture

Which Gene Variants Drive FH?

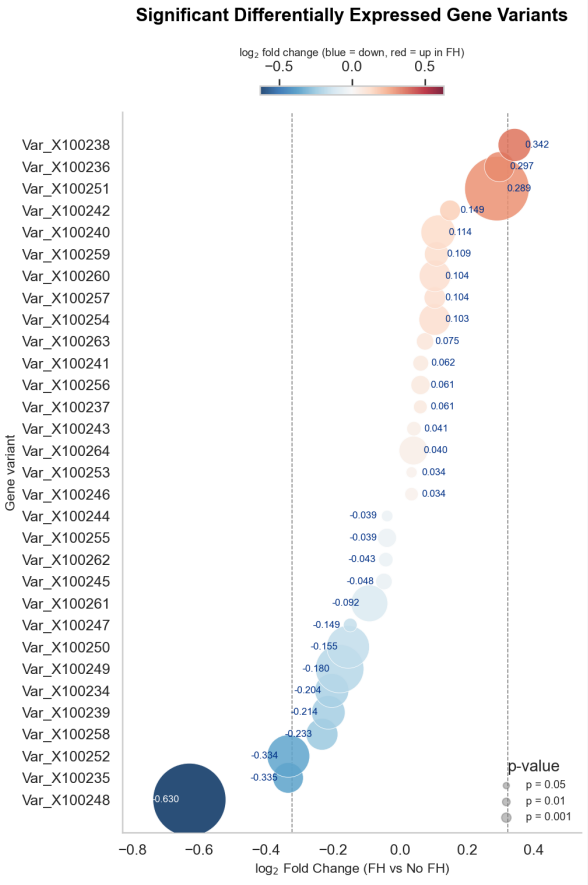


Key findings

No single variant exceeds ±1.0 $\log_2$FC i.e no dominant gene signal

Signal distributed across many small effects

Which Gene Variants Drive FH?



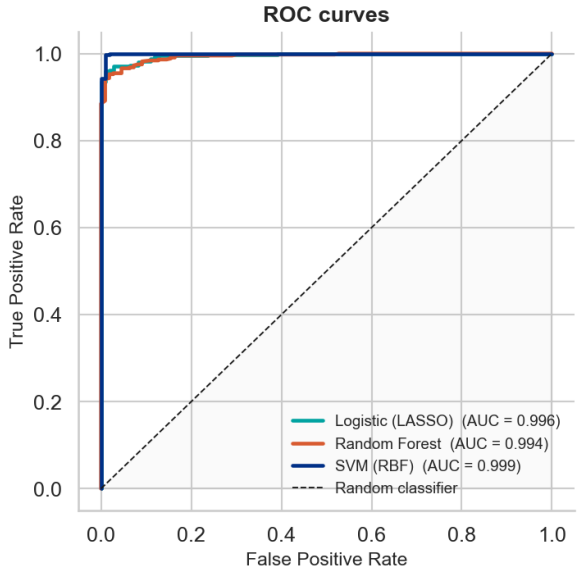
Key findings

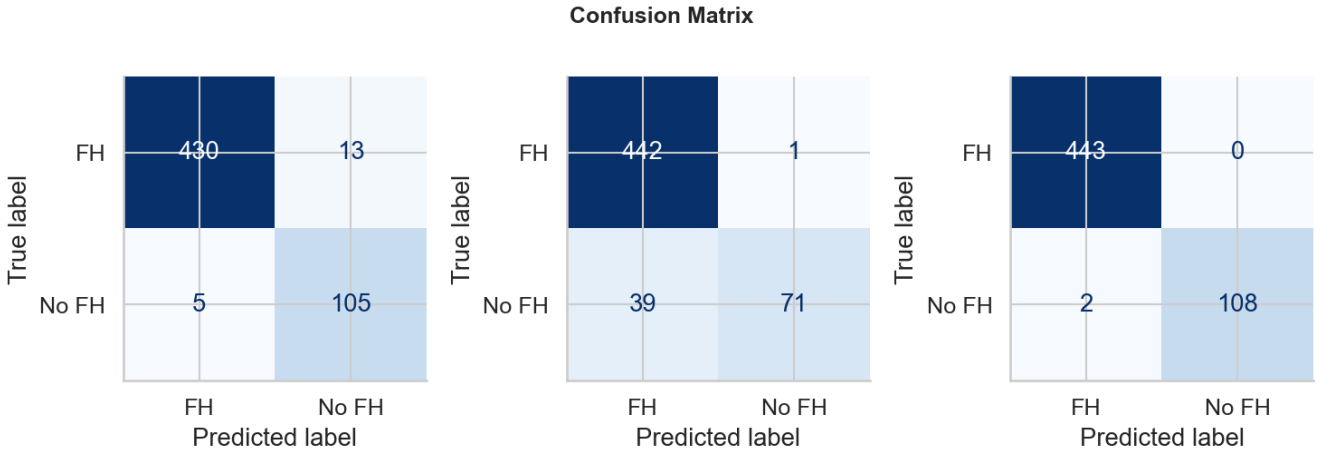
Var_X100248 — most strongly & significantly downregulated in FH ($\log_2$FC = –0.630, ~54% lower expression)

Var_X100251 — most statistically significant upregulated variant (largest bubble on right)

Var_X100238 — largest upregulated fold change ($\log_2$FC = +0.342, ~27% higher expression)

Can we predict FH?





0.996

AUC — test set

Logistic (LASSO)

FH Recall	97.1%
Specificity	95.5%
Precision	98.9%

0.994

AUC — test set

Random Forest

FH Recall	99.8%
Specificity	64.5%
Precision	91.9%

Recommended

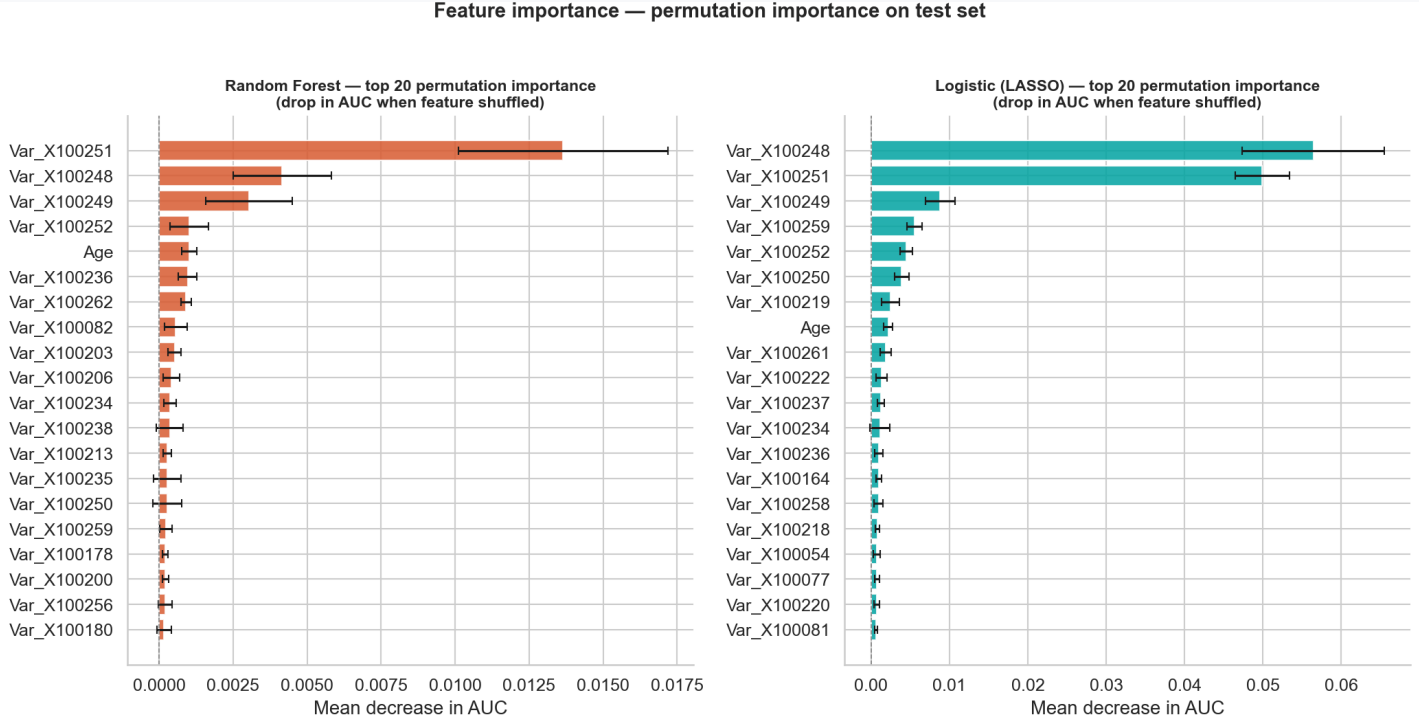
0.999

AUC — test set

SVM (RBF kernel)

FH Recall	100.0%
Specificity	98.2%
Precision	99.6%

Do the Prediction Models Agree on Which Variants Matter?



Var_X100248 (downregulated) and **Var_X100251** (upregulated) are the most influential features — consistent across Random Forest, LASSO, and volcano plot

Limitations

Current dataset limitations

Gene expression variability and cross-sectional design

expression levels fluctuate with medication, fasting status and time of day; without standardised collection protocols and longitudinal follow-up

Class imbalance

80% FH vs 20% control — far from real-world prevalence (~0.4%). Models need recalibration before clinical use.

No lipid panel values

LDL-C, HDL-C and total cholesterol are the primary diagnostic markers for FH.